9 Synthetic-lethal vulnerability map (Ruppin-style) STRATEGIC PLAN

DM1-anchored thyroid SL/SR — design-only forward-look paper. Inherits authority from Paper 1 once it is in print.

Status: STRATEGIC DESIGN ONLY · no data download · no analysis · no GPU · no manuscript text. **Earliest start: 2026-06-16** (after marathon close 2026-06-13). Anchored on Paper 1 (DM1 dark matter) — Paper 9 cannot start until Paper 1 is in print.

QUESTION

Given the DM1 / RAI-lineage-collapsed state, what genes / drugs kill the tumor selectively, and which compensatory partners must we co-target?

ANCHOR

8-gene compact readout + TF-collapse score (FOXE1 / NKX2-1 / PAX8 / HHEX) — state-anchored, not mutation-anchored

CONCEPTUAL FRAME

Ruppin-style SL/SR (concepts only — no text, no precomputed pair lists lifted) · thyroid-tuned · DM1-conditioned

DATA LATER

DepMap · CCLE · PRISM · GDSC · CTRP · TCGA-THCA · GSE76039 · thyroid PDX/organoid panels

EARLIEST START

2026-06-16 (post-marathon)

STOP CONDITIONS

Step A (DM1-like state in CCLE) fail → reframe; Step C (SL pairs) zero hits → publish as ranked dependency list in lower-tier venue

Hypothesis decomposition

H	CLAIM	TESTED WHERE
H0	DM1-like thyroid tumors carry a non-redundant set of synthetic-lethal vulnerabilities arising from lineage collapse + compensatory rewiring; predictable from 8-gene readout without driver mutation profiling	Whole paper
H1	DM1-like state is reproducibly callable in CCLE	Step A
H2	A subset of CRISPR dependencies is enriched in DM1-like lines	Step B (DepMap)
H3	A subset has synthetic-rescue partners predicting resistance	Step C
H4	Drug response in PRISM/GDSC/CTRP recapitulates H2 for druggable targets	Step D
H5	High target-pathway expression in DM1-classified tumors stratifies survival	Step E (TCGA-THCA + GSE76039)

Candidate target classes

SL · NAD SALVAGE

NAMPT

STAT3-high → NAD turnover. SR partner: NAPRT (de novo arm). Drugs: FK866, KPT-9274, OT-82. Falsifier: NAPRT uniformly high in DM1.

SL · CYTOKINE / TF

JAK / STAT3

Direct DM1 inflammatory readout. Real SL likely upstream (IL6R, OSMR) or parallel TFs (NFκB). Drugs: ruxolitinib, deucravacitinib-class.

REDIFFERENTIATION

DNMT

Hypermethylation of FOXE1/PAX8/NKX2-1/HHEX. Lineage rewiring, not pure SL. Drugs: decitabine, guadecitabine ± HDACi.

SL · COMPENSATORY KINASE

SFK (LYN, FYN, SRC)

Compensatory tyrosine-kinase signaling in mutation-poor dedifferentiated thyroid. Drugs: dasatinib, bosutinib, saracatinib.

MODIFIER

KCNN4 (KCa3.1)

Calcium-activated K+ channel; immune-tumor crosstalk. Drugs: senicapoc, TRAM-34. Likely modifier rather than hard SL.

SL · DNA DAMAGE

PARP / ATR

Lineage plasticity correlates with replication stress. Drugs: olaparib, talazoparib, ceralasertib. SR: HR-intact resistance.

SL · METABOLIC

OXPHOS / glutamine

Lineage collapse forces metabolic reprogramming. Drugs: IACS-010759 (OXPHOS), CB-839 (glutaminase). Pair: GLS1 ↔ MYC.

⚠️ NON-SL — CARVED OUT

TROP2 (ADC)

Tumor surface marker, NOT a synthetic-lethal partner. Sacituzumab govitecan = expression-level vulnerability (parallel translational opportunity). Mixing with SL is a recurring failure mode — explicitly carved out.

12-week post-marathon execution plan

Floor: 2026-06-13 (manuscript marathon close). Earliest practical start: **2026-06-16**. Each week = checkpoint, not deadline.

WK	PHASE	OUTPUT
1	DepMap + CCLE pull, integrity checks	Local mirror, version-pinned
2	Define DM1-like state in CCLE; score thyroid lines	DM1 score table, cell-line ranking
3	Genome-wide DM1-conditioned dependency analysis	Volcano (F3 draft)
4	SL / SR pair inference + Ruppin sanity cross-check	Pair table (F4 draft)
5	PRISM / GDSC pull and integration	Drug-response matrix
6	Drug-response stratification by DM1	Per-target IC50 panels (F5 draft)
7	TCGA-THCA + GSE76039 patient validation	Survival forest plots (F6 draft)
8	Cross-checks: lineage, batch, confounders	Sensitivity tables
9	Figure assembly F1–F6, supplements	Figure pack v1
10	Draft writing (intro, results, discussion)	Manuscript v0
11	Internal review with Yu / collaborator(s); revise	Manuscript v1

WK	PHASE	OUTPUT
12	Submission package	Submitted

What not to claim (hard rules)

❌ Direct equivalence with ISLE/SELECT outputs – conceptually inspired only

❌ Clinical efficacy from cohort survival stratification – hypothesis-generating only

❌ DM1-readout replaces driver-mutation profiling – frame as *complementary*

❌ Mix TROP2-ADC vulnerability with synthetic lethality – different mechanistic categories (Sec 6.8)

❌ Over-extend to Hashimoto-overlap or Graves' subtypes – Paper 2 / Paper 4 territory

❌ Use the H&E-DM1 angle – NO-GO per existing standing decision

❌ Wet-lab validation we have not done – explicit "computational + cohort-validated, wet-lab pending"

❌ Lift sentences / signature lists / precomputed pair tables from Ruppin-group papers – concept yes, text/data no

❌ Start any of this before Paper 1 in print + marathon closed (2026-06-13 floor)

IP ANGLE

Method-of-use / companion-diagnostic IP slot: "method of selecting thyroid cancer patients for [target-class] therapy by measuring the 8-gene DM1 readout + TF-collapse score". 8-gene readout = most defensible asset.

TRIAL DESIGN

Umbrella / basket trial routing patients by DM1 score to one of 3-4 target arms (NAD / JAK-STAT / epigenetic / ADC). Pitchable to industry partners after publication.

WET-LAB TIER

Tier 1 (defendable): siRNA/CRISPR + drug-response in 2-3 thyroid lines for top 2 SL targets. Tier 2 (stretch): organoid + combination. Tier 3 (follow-up): PDX in vivo.

FALLBACK VENUE

If no wet-lab partner: still publishable as computational + patient-cohort vulnerability map (JCI Insight / Mol Cancer Therap / NPJ Precision Oncology) – hypothesis-generating frame.

Reference

Paper 9 는 strategic plan only – manuscript / PDF / figure 가 아직 없습니다. 전체 14-section plan 은 아래 markdown 에서 보세요.

strategic plan (md, 307 lines) Paper 1 (DM1 anchor) Phase B drug platform

legacy trajectory

strategic plan (md) Paper 1 anchor Phase B platform ← hub